

StarPay Matrix

Mutual Fund TER Portal
Complete Project Documentation

Functional, technical and operational reference

Prepared from the implemented source code and runtime verification

Runtime verification date: 19 September 2026

Mutual Fund TER Portal

1. Overview

The Mutual Fund TER Portal is an Express/EJS web application for viewing and maintaining mutual-fund Total Expense Ratio (TER), NAV, brokerage, commission, launch-date, and AMC document data.

The application fetches TER data and NAV data from AMFI, matches the two datasets by normalized scheme name, and serves the result through a searchable matrix UI and JSON APIs. Operational data is persisted as JSON files and AMC document folders. User accounts are stored in SQLite.

2. Technology Stack

Area	Technology
Runtime	Node.js <code>>=22.5.0</code>
Server	Express 4
Views	EJS
Browser UI	Vanilla JavaScript and CSS
External sources	AMFI TER endpoints and <code>NAVA11.txt</code>
Excel files	<code>xlsx</code>
Authentication	HMAC-signed cookie sessions and <code>bcryptjs</code>
User database	Node built-in SQLite (<code>node:sqlite</code>)
Scheduling	<code>node-schedule</code> , timezone <code>Asia/Kolkata</code>
Security	Helmet, CSRF double-submit tokens, login rate limiting, upload validation

3. Quick Start

Prerequisites

- Node.js `22.5.0` or newer
- npm
- Network access to AMFI for a live refresh

Install and run

```
cd "c:\Users\ss\Desktop\New folder (10)\ter-portal"
npm install
npm start
```

Open the URL printed in the server log. The default HTTP port is `3000`. If that port is already in use, the server tries the next five ports automatically.

Development mode, with automatic restart:

```
npm run dev
```

Run automated tests:

```
npm test
```

4. Configuration

Configuration is read from environment variables by `config/app.js` and the services that own each setting. A local `.env` file is supported through `dotenv`.

Variable	Default	Purpose
<code>PORT</code>	<code>3000</code> (<code>7443</code> with HTTPS)	Listening port
<code>HOST</code>	<code>0.0.0.0</code>	Listening address
<code>NODE_ENV</code>	<code>development</code>	Runtime environment
<code>HTTPS</code>	<code>false</code>	Enable HTTPS
<code>TLS_PFX_PATH</code>	<code>certs/lan.pfx</code>	PFX certificate path
<code>TLS_PFX_PASSPHRASE</code>	<code>unset</code>	PFX passphrase
<code>SESSION_SECRET</code>	local development fallback	Session HMAC secret; required in production
<code>ADMIN_USERNAME</code>	<code>admin</code>	Seed username when the user store is empty
<code>ADMIN_PASSWORD</code>	<code>admin123</code>	Seed password when the user store is empty
<code>CACHE_REFRESH_INTERVAL</code>	<code>30</code>	Legacy/general cache interval setting
<code>REQUEST_TIMEOUT</code>	<code>30000</code>	AMFI request timeout in milliseconds
<code>AMFI_TER_URL</code>	AMFI TER page	TER data source
<code>AMFI_NAV_URL</code>	AMFI <code>NAVA11.txt</code>	NAV data source

Variable	Default	Purpose
DB_PATH	data/app.db	SQLite database path
TER_CACHE_PATH	ter-cache.json	Main scheme cache path
TER_MANUAL_SYNC_PATH	ter-manual-sync.json	Manual sync snapshot path
COMMISSION_OVERRIDE_CACHE_PATH	commission-override-cache.json	Commission override path
LAUNCH_DATE_OVERRIDE_CACHE_PATH	launch-date-override-cache.json	Launch-date override path
LAUNCH_DATE_WORKBOOK_PATH	configured reference workbook	Launch-date workbook path
AMC_DOCUMENTS_ROOT	data/amc-documents	AMC document root
USERS_FILE_PATH	data/users.json	One-time legacy user import source

For production, set a long random `SESSION_SECRET`, change the default admin password, and use HTTPS with a valid certificate.

5. Application Flow

```

flowchart TD
    Browser --> Express
    Express --> Security[Helmet, CSRF, login rate limit]
    Security --> Routes[Page and API routes]
    Routes --> Controllers
    Controllers --> Services
    Services --> Cache[In-memory cache]
    Services --> AMFI[AMFI TER and NAV sources]
    Cache --> JSON[JSON cache and override files]
    Services --> SQLite[(SQLite users)]
    Services --> Docs[AMC document folders]

```

At startup, the server loads environment configuration, mounts middleware and routes, loads persisted cache/overrides, and starts the scheduler. If a usable cache exists, it is served immediately and a refresh runs in the background. Without usable cache data, startup performs a refresh before the scheduler starts.

The full refresh fetches TER and NAV data in parallel, merges them using `services/matchService.js`, validates the resulting size, and persists a good result. A suspiciously small response is rejected so an AMFI failure cannot overwrite a known-good cache. A refresh failure leaves existing cached data available.

6. Scheduled Refresh

The scheduler in `services/schedulerService.js` runs a full AMFI refresh daily at:

- 10:00 IST
- 12:00 IST
- 14:00 IST
- 16:00 IST

The schedule uses `Asia/Kolkata`, not the machine's local timezone. A manual refresh is also available to authenticated `admin` and `data_operator` users through `GET /api/refresh`.

NAV data is reused in memory for four hours. Period-specific TER requests use a 25-second timeout and fall back to the last available data with a stale marker when AMFI is unavailable.

7. User Interface

Page	URL	Purpose
TER matrix	<code>/</code>	Browse, filter, sort, edit, and export current scheme data
Old brokerage	<code>/old-brokerage-data</code>	View and maintain historical brokerage records
Login	<code>/login</code>	Authenticate a user
User administration	<code>/admin/users</code>	Create, update, and delete users; admin only
About	<code>/about</code>	Application information
AMC shortcut	<code>/:amc</code>	Render the home view for an AMC route

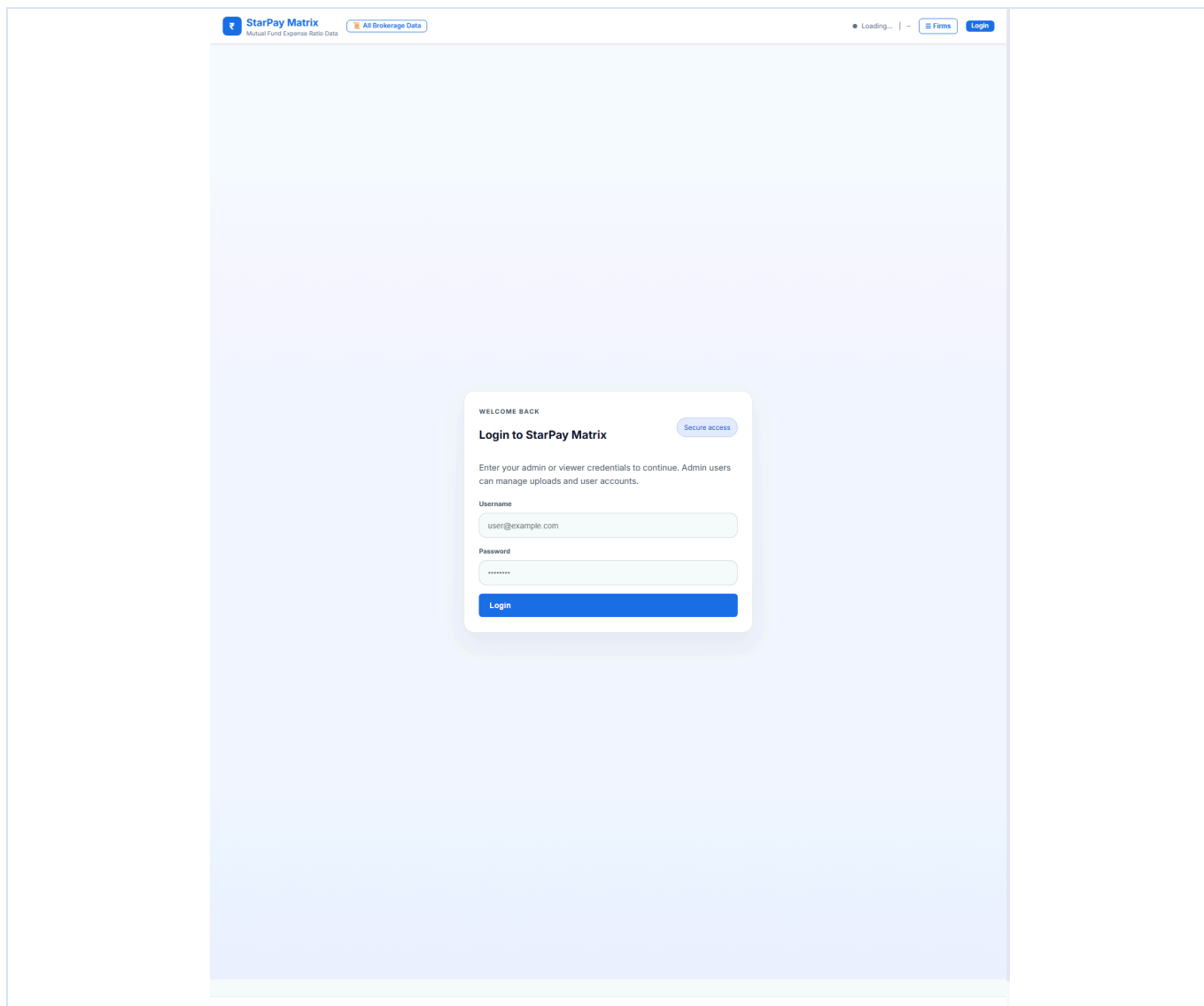
The main matrix supports AMC, category, type, scheme, NSDL, and AMFI filters, pagination, column visibility, CSV/Excel export, clipboard copy, printing, and role-based editing. The old brokerage page supports validity-period filtering, historical uploads, editing, deletion, and exports.

8. Step-by-Step Page Guide with Screenshots

The following screenshots show the main user-facing pages in the normal workflow. Screenshots are stored in `docs/screenshots/` and are embedded in the PDF version of this document.

Step 1: Login

1. Open `/login`.
2. Enter the username and password supplied by an administrator.
3. Select **Login**.
4. After authentication, return to the portal home page. Admin and data-operator users see additional upload and management actions.



Login page

Step 2: Review the TER Matrix

1. Open [/](#).
2. Review the dashboard metrics and latest data period.
3. Select an AMC, category, type, brokerage period, or search field to filter schemes.
4. Use column sorting, pagination, export, copy, or print controls as needed.
5. Authenticated data-role users can select a row's edit action to update permitted values.

StarPay Matrix

Mutual Fund Expense Ratio Data

All Brokerage Data

Live

Updated: 19/09/2026, 11:29:09 am

Firms

Login

StarPay Matrix

Live Total Expense Ratio data for all AMFI registered schemes

2,137

Total Schemes

53

AMC's

1 Jul 2026 – 30 Sep 2026

Latest Data Period

Live

Data Status

Data Sync Backup

Save a known-good copy before refreshing data. The saved copy can be opened if live scraping fails.

Get Old Sync Data

Restricted actions

Login as admin to upload Excel files and manage users. Regular viewers can browse the latest TER data.

Filter Schemes

AMC

All AMCs

View/Manage Documents

SCHEME CATEGORY

All Categories

SCHEME TYPE

All Types

BROKERAGE DATA

All Schemes

SCHEME NAME

Search scheme name...

NSDL CODE

AMFI CODE

e.g. INF123...

e.g. 100016

X Reset Filters

Showing 2,137 of 2,137 schemes

Columns

Rows: 50

Export All

CSV

Excel

Copy

Print

#	NSDL CODE	AMFI CODE	SCHEME NAME	LAUNCH DATE	AMC	SCHEME TYPE	CATEGORY	DATE PERIOD
1	3680/O/I/BH/23/07/0007	152873	360 ONE Balanced Hybrid Fund 360 ONE Mutual Fund	24 Sep 2023	360 ONE Mutual Fund	Open Ended	Hybrid Schemes - Balanced Hybrid Fund	1st July 2026 to 30th September 2026
2	3680/O/I/DB/13/02/0001	122612	360 ONE Dynamic Term Fund 360 ONE Mutual Fund	-	360 ONE Mutual Fund	Open Ended	Income/Debt Oriented Schemes - Dynamic Term Fund	-
3	3680/O/I/EL/22/10/0005	151165	360 ONE ELSS Tax Saver Nifty 50 Index Fund 360 ONE Mutual Fund	27 Dec 2022	360 ONE Mutual Fund	Open Ended	Index Funds - Equity Funds	1st July 2026 to 30th September 2026
4	3680/O/I/FC/23/02/0006	151796	360 ONE FLEXICAP FUND 360 ONE Mutual Fund	29 Jun 2023	360 ONE Mutual Fund	Open Ended	Equity Schemes - Flexi Cap Fund	1st July 2026 to 30th September 2026
5	3680/O/I/FO/24/06/0003	111583	360 ONE Focused Fund 360 ONE Mutual Fund	29 Oct 2014	360 ONE Mutual Fund	Open Ended	Equity Schemes - Focused Fund	-
6	3680/O/I/GT/25/01/0000	153357	360 ONE GOLD ETF 360 ONE Mutual Fund	06 Mar 2025	360 ONE Mutual Fund	Open Ended	Exchange Traded Funds (ETFs) - Gold ETF	-
7	3680/O/I/LI/23/05/0002	125349	360 ONE Liquid Fund 360 ONE Mutual Fund	12 Nov 2013	360 ONE Mutual Fund	Open Ended	Income/Debt Oriented Schemes - Liquid Fund	1st July 2026 to 30th September 2026
8	3680/O/I/MI/26/04/0012	154586	360 ONE MSCI India ETF 360 ONE Mutual Fund	21 May 2026	360 ONE Mutual Fund	Open Ended	Exchange Traded Funds (ETFs) - Other ETF	-
9	3680/O/I/M/25/04/0011	153774	360 ONE Multi Asset Allocation Fund 360 ONE Mutual Fund	19 Aug 2025	360 ONE Mutual Fund	Open Ended	Hybrid Schemes - Multi Asset Allocation Fund	1st July 2026 to 30th September 2026
10	3680/O/I/O/25/01/0018	153783	360 ONE OVERNIGHT FUND 360 ONE Mutual Fund	09 Jul 2025	360 ONE Mutual Fund	Open Ended	Income/Debt Oriented Schemes - Overnight Fund	1st July 2026 to 30th September 2026
11	3680/O/I/TH/23/10/0004	149318	360 ONE QUANT FUND 360 ONE Mutual Fund	28 Nov 2021	360 ONE Mutual Fund	Open Ended	Equity Schemes - Thematic Fund	1st July 2026 to 30th September 2026
12	3680/O/I/ST/25/01/0000	153415	360 ONE SILVER ETF 360 ONE Mutual Fund	27 Mar 2025	360 ONE Mutual Fund	Open Ended	Exchange Traded Funds (ETFs) - Silver ETF	-
13	ABKS/O/I/FC/25/11/0002	154843	Abakus Flexi Cap Fund Abakus Mutual Fund	28 Dec 2025	Abakus Mutual Fund	Open Ended	Equity Scheme - Flexi Cap Fund	Q2 - July August September - 2026
14	ABKS/O/I/LI/26/06/0005	154477	Abakus Large & Mid Cap Fund Abakus Mutual Fund	-	Abakus Mutual Fund	Open Ended	Equity Scheme - Large & Mid Cap Fund	-
15	ABKS/O/I/LI/25/11/0001	154849	Abakus Liquid Fund Abakus Mutual Fund	14 Dec 2025	Abakus Mutual Fund	Open Ended	Income/Debt Oriented Schemes - Liquid Fund	Q2 - July August September - 2026
16	ABKS/O/I/SC/26/01/0003	154215	Abakus Small Cap Fund Abakus Mutual Fund	16 Mar 2026	Abakus Mutual Fund	Open Ended	Equity Scheme - Small Cap Fund	Q2 - July August September - 2026
17	ABSL/O/I/BH/25/01/0001	128517	Aditya Birla Sun Life Aggressive Hybrid Fund Aditya Birla Sun Life Mutual Fund	-	Aditya Birla Sun Life Mutual Fund	Open Ended	Hybrid Schemes - Aggressive Hybrid Fund	-
18	ABSL/O/I/FO/28/10/0042	131275	Aditya Birla Sun Life Aggressive Hybrid Omni FOF	08 Mar 2011	Aditya Birla Sun Life Mutual	Open Ended	Other Scheme - FoF	01-Jul-2026 to 30-Sep-2026

TER matrix home page

Step 3: Review Historical Brokerage

1. Select **All Brokerage Data** or open /old-brokerage-data .
2. Filter by financial year, date range, AMC, category, ARN, or scheme name.
3. Review the validity period and year-wise brokerage values.
4. Use export or print controls for reporting.
5. Authenticated data-role users can upload, add, edit, or delete historical records.

Old Brokerage Data

Date-wise historical distributor trail/brokerage structures — sourced from Mutual_Fund_Brokerage_Consolidated_Verified.xlsx

6,681
Total Records58
AMCs2023-10-01 -- 2026-09-30
Data Available Range

Filter Old Brokerage Data

X Reset Filters

FINANCIAL YEAR	FROM DATE	TO DATE	AMC	CATEGORY / TIER	ARN NO.
All Time	dd-mm-yyyy	dd-mm-yyyy	All AMCs	All Categories	All ARN Nos.
SCHEME NAME		BROKERAGE DATA			
Search scheme name...		All Schemes			

Tip: the data filter matches each scheme's source validity period against the range you pick — rows whose validity window overlaps "From — To" are shown. Records still active with no fixed end date ("onwards / till further notice") are treated as valid through today.

Note: BER (%) and C/8 1-6YR are calculated automatically for every row — they always use that scheme's current BER% from the main StarPay Matrix table (BER doesn't change per old quarter, only the commission % does), so they stay in sync automatically and can't be edited here.

Showing 6,681 of 6,681 records

Columns

Rows: 50

Export All

CSV

Copy

Print

#	AMC / FIRM	SCHEME NAME	CATEGORY / TIER	ARN NO.	DATA PERIOD (VALIDITY)	1ST YR %	2ND YR %	3RD YR %	4TH YR %	5Y
1	360 ONE Mutual Fund	360 ONE Balanced Hybrid Fund	Hybrid	13025	1st April 2026 – 30th June 2026	0.8%	0.8%	0.8%	0.8%	
2	360 ONE Mutual Fund	360 ONE Dynamic Bond Fund	Debt	13025	1st April 2026 – 30th June 2026	0.21%	0.21%	0.21%	0.21%	
3	360 ONE Mutual Fund	360 ONE ELSS Nifty 50 Tax Saver Index Fund	Equity	13025	1st April 2026 – 30th June 2026	0.21%	0.21%	0.21%	0.21%	
4	360 ONE Mutual Fund	360 ONE Flexicap Fund	Equity	13025	1st April 2026 – 30th June 2026	0.75%	0.75%	0.75%	0.75%	
5	360 ONE Mutual Fund	360 ONE Focused Equity Fund	Equity	13025	1st April 2026 – 30th June 2026	0.6%	0.6%	0.6%	0.6%	
6	360 ONE Mutual Fund	360 ONE Liquid Fund	Liquid	13025	1st April 2026 – 30th June 2026	0.04%	0.04%	0.04%	0.04%	
7	360 ONE Mutual Fund	360 ONE Multi Asset Allocation Fund	Hybrid	13025	1st April 2026 – 30th June 2026	1%	1%	1%	1%	
8	360 ONE Mutual Fund	360 ONE Overnight Fund	Debt	13025	1st April 2026 – 30th June 2026	0.04%	0.04%	0.04%	0.04%	
9	360 ONE Mutual Fund	360 ONE Quant Fund	Equity	13025	1st April 2026 – 30th June 2026	0.85%	0.85%	0.85%	0.85%	
10	Abakus Mutual Fund	Abakus Flexi Cap Fund	Equity	280532	1 Jan 2026 – 31 Mar 2026	1.2%	1.2%	1.2%	1.1%	
11	Abakus Mutual Fund	Abakus Flexi Cap Fund	–	280532	1st April 2026 – 30th June 2026	1.2%	1.2%	1.2%	1.1%	
12	Abakus Mutual Fund	Abakus Liquid Fund	Liquid	280532	1 Jan 2026 – 31 Mar 2026	0.08%	0.08%	0.08%	0.08%	
13	Abakus Mutual Fund	Abakus Liquid Fund	–	280532	1st April 2026 – 30th June 2026	0.08%	0.08%	0.08%	0.08%	
14	Abakus Mutual Fund	Abakus Small Cap Fund	–	280532	1st April 2026 – 30th June 2026	1.4%	1.4%	1.4%	1.3%	
15	Abakus Mutual Fund	Abakus Small Cap Fund	Equity	280532	26 Feb 2026 – 12 Mar 2026	1.4%	1.4%	1.4%	1.3%	
16	Aditya Birla Sun Life Mutual Fund	ADITYA BIRLA SUN LIFE NIFTY 50 EQUAL WEIGHT INDEX FUND	INDEX	280532	01-Apr-2026 to 30-Jun-2026	0.45%	0.45%	0.45%	0.45%	
17	Aditya Birla Sun Life Mutual Fund	ADITYA BIRLA SUN LIFE AGGRESSIVE HYBRID OMNI FOF	EQUITY - 2	280532	01-Apr-2026 to 30-Jun-2026	0.45%	0.45%	0.45%	0.45%	
18	Aditya Birla Sun Life Mutual Fund	ADITYA BIRLA SUN LIFE ARBITRAGE FUND	ARBITRAGE	280532	01-Apr-2026 to 30-Jun-2026	0.45%	0.45%	0.45%	0.45%	
19	Aditya Birla Sun Life Mutual Fund	ADITYA BIRLA SUN LIFE ARBITRAGE FUND	ARBITRAGE	280532	01-Apr-2026 to 31-May-2026	0.45%	0.45%	0.45%	0.45%	
20	Aditya Birla Sun Life Mutual Fund	ADITYA BIRLA SUN LIFE ARBITRAGE FUND	ARBITRAGE	280532	01-Jun-2026 to 30-Jun-2026	0.7%	0.55%	0.45%	0.45%	
21	Aditya Birla Sun Life Mutual Fund	ADITYA BIRLA SUN LIFE BAL BHAVISHYA YOJANA WEALTH PLAN	SOLUTION BASED	280532	01-Apr-2026 to 30-Jun-2026	0.95%	0.95%	0.95%	0.95%	
22	Aditya Birla Sun Life Mutual Fund	ADITYA BIRLA SUN LIFE BAL BHAVISHYA YOJANA WEALTH PLAN	SOLUTION BASED	280532	FUND01-Apr-2026 to 30-Jun-2026 X FUND	0.95%	0.95%	0.95%	0.95%	
23	Aditya Birla Sun Life Mutual Fund	ADITYA BIRLA SUN LIFE BALANCED ADVANTAGE FUND	HYBRID	280532	01-Apr-2026 to 30-Jun-2026	0.75%	0.75%	0.75%	0.75%	
24	Aditya Birla Sun Life Mutual Fund	ADITYA BIRLA SUN LIFE BALANCED ADVANTAGE FUND	HYBRID	280532	01-May-2026 to 31-May-2026	0.4%	0.4%	0.4%	0.4%	
25	Aditya Birla Sun Life Mutual Fund	ADITYA BIRLA SUN LIFE BANKING & PSU DEBT FUND	LT DEBT	280532	01-Apr-2026 to 30-Jun-2026	0.3%	0.3%	0.3%	0.3%	
26	Aditya Birla Sun Life Mutual Fund	ADITYA BIRLA SUN LIFE BANKING & PSU DEBT FUND	LT DEBT	280532	FUND X FUND 01-Apr-2026 to 30-Jun-2026	0.3%	0.3%	0.3%	0.3%	
27	Aditya Birla Sun Life Mutual Fund	ADITYA BIRLA SUN LIFE BANKING AND FINANCIAL SERVICES FUND	SECTORAL/THEMATIC	280532	01-Apr-2026 to 30-Jun-2026	0.8%	0.8%	0.8%	0.8%	

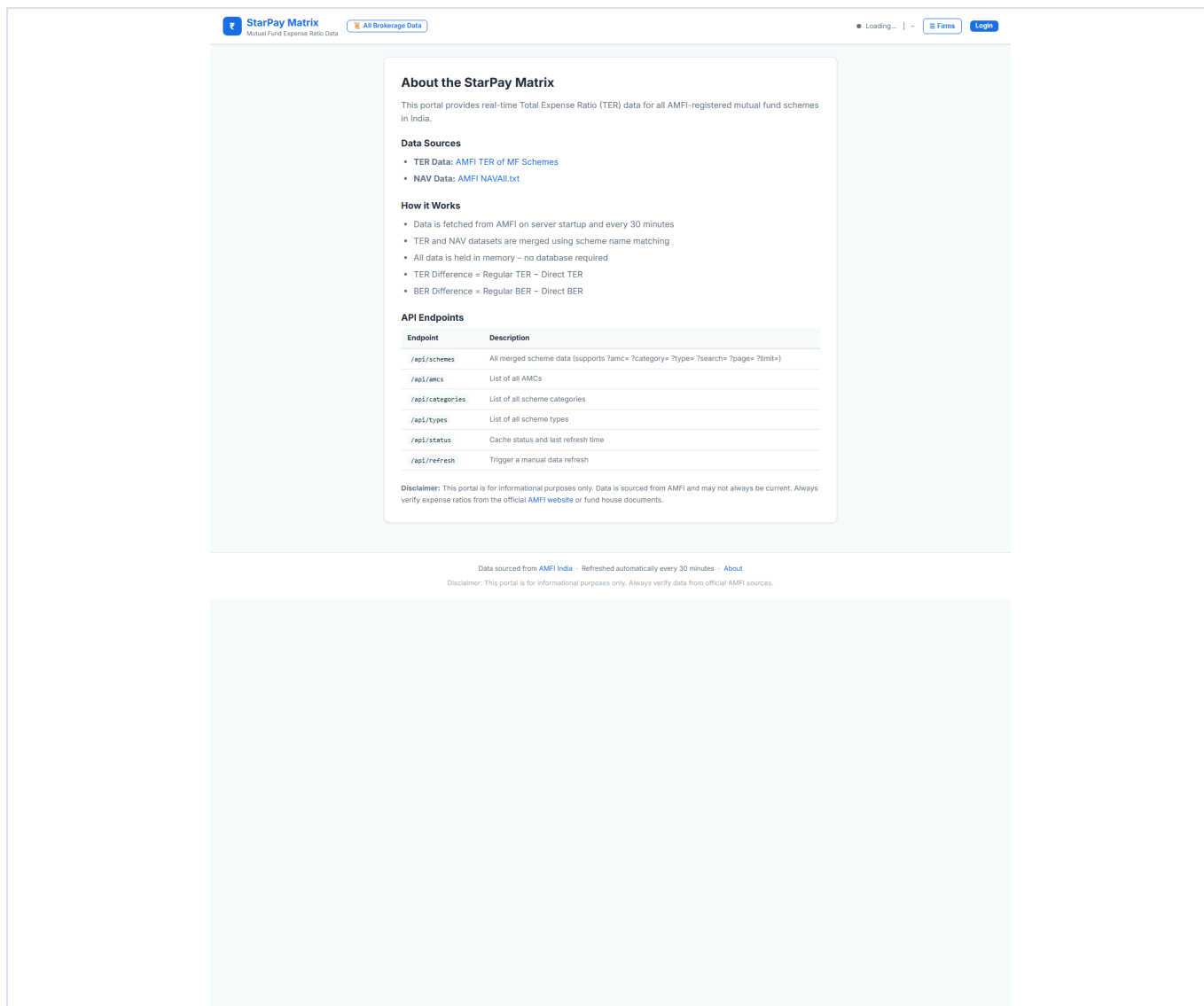
Aditya Birla Sun Life Mutual Fund

ADITYA BIRLA SUN LIFE BANKING AND FINANCIAL SERVICES FUND

Old brokerage page

Step 4: View About and Data Source Information

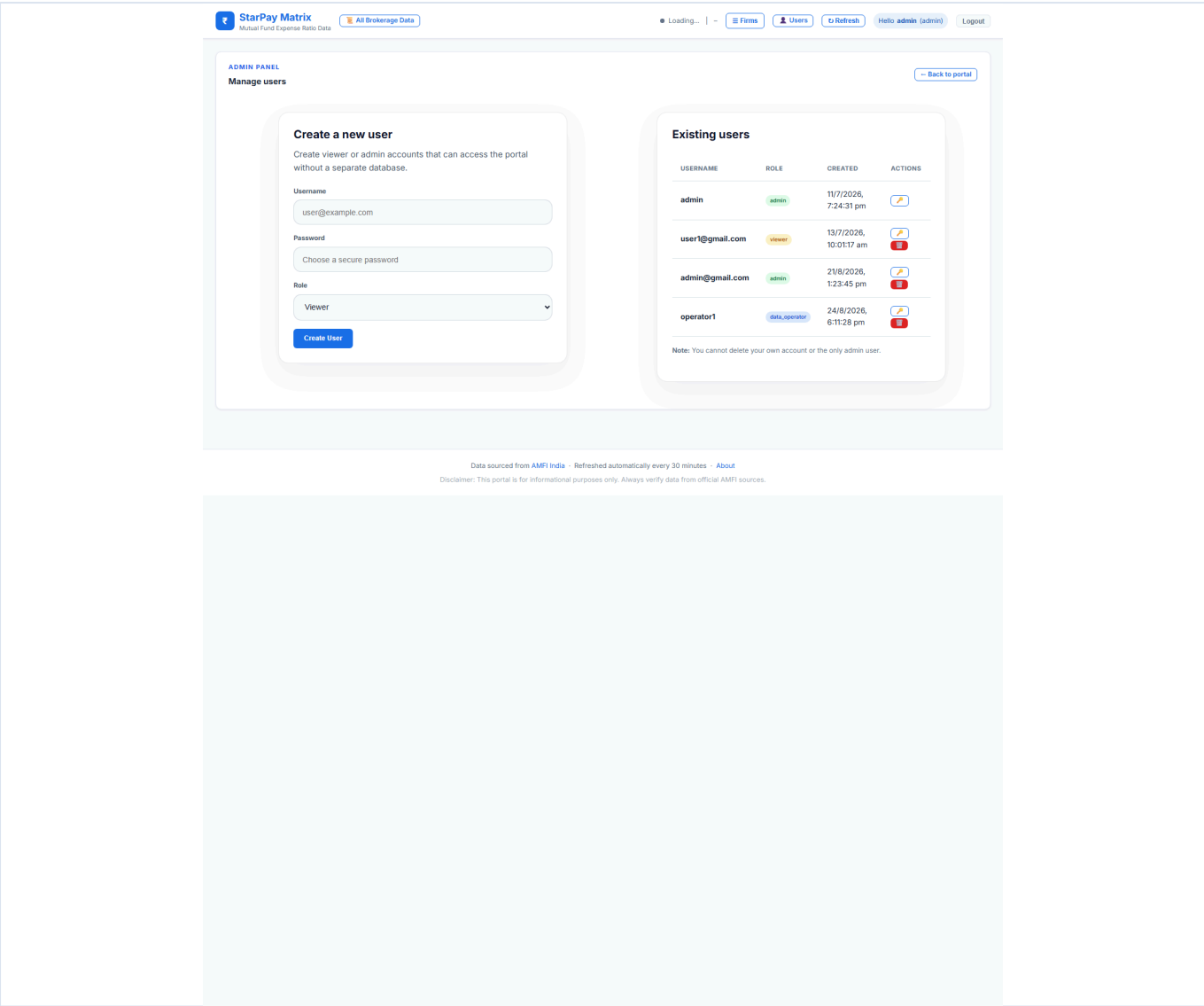
1. Select **About** from the footer or open /about .
2. Review the AMFI data sources, matching behavior, calculations, API summary, and disclaimer.



About page

Step 5: Manage Users as an Administrator

1. Sign in with an administrator account.
2. Open **Users** or `/admin/users`.
3. Create a viewer, data-operator, or admin account by entering a username, password, and role.
4. Use the key action to change a user's password.
5. Use the delete action when an account is no longer required.
6. The application prevents self-deletion and deletion of the only remaining admin account.



Admin user management page

9. Roles and Permissions

Valid roles are `admin`, `data_operator`, and `viewer`.

Capability	Viewer	Data operator	Admin
View, filter, and export data	Yes	Yes	Yes
Edit scheme and launch-date values	No	Yes	Yes
Upload commission or brokerage data	No	Yes	Yes
Save sync and clear overrides	No	Yes	Yes
Manage AMC documents	No	Yes	Yes
Manage users	No	No	Yes

The only-admin safeguards prevent changing or deleting the final administrator. Usernames are normalized to lowercase and passwords must contain at least four characters. Passwords are stored as bcrypt hashes. Legacy

SHA-256 records are upgraded to bcrypt after a successful login.

10. Data Sources and Matching

AMFI TER and NAV

`services/terService.js` retrieves TER periods and rows from AMFI. `services/navService.js` retrieves and parses the configured NAV text file. `services/matchService.js` normalizes scheme names and computes a similarity score. The configured minimum match confidence is `80`.

Unmatched TER rows remain visible; they simply do not receive a matched AMFI/NAV code or value.

Launch dates

Launch-date resolution uses this priority:

1. Manual override in `launch-date-override-cache.json`.
2. Reference workbook configured by `LAUNCH_DATE_WORKBOOK_PATH`.
3. Existing AMFI/cache value.
4. Empty value.

Commission and brokerage

Commission uploads and inline edits are persisted in `commission-override-cache.json`. Historical brokerage changes are persisted in `old-brokerage-override-cache.json`, including deletion tombstones. Old brokerage records are keyed by AMC, scheme, and period, and only supported ARN values are exposed by the service.

For each year, the commission-to-BER ratio is:

$$C/B_n = \frac{\text{Commission}_n}{\text{Regular BER}}$$

The UI displays a dash when commission or BER is missing, or when BER is zero.

11. API Reference

All API paths below are relative to `/api`.

Read endpoints

Method	Endpoint	Description
GET	<code>/schemes</code>	Scheme matrix; supports period, filter, paging, and search parameters
GET	<code>/amcs</code>	AMC catalog
GET	<code>/categories</code>	Scheme categories
GET	<code>/types</code>	Scheme types
GET	<code>/status</code>	Cache status and refresh information

Method	Endpoint	Description
GET	/dashboard	Dashboard counts and TER statistics
GET	/periods	Available financial years
GET	/months?year=...	Available TER months for a year
GET	/old-brokerage	Historical brokerage data
GET	/sync/old	Saved manual sync snapshot
GET	/amcs/:amc/documents	List AMC documents
GET	/amcs/:amc/documents/:documentId/file	View/download a document
GET	/amcs/:amc/documents/:documentId/download	View/download a document

Protected data endpoints

These require an authenticated `admin` or `data_operator` session unless noted otherwise.

Method	Endpoint	Description
GET	/refresh	Start a background full refresh
POST	/sync/save	Save the current cache as a manual snapshot
POST	/clear-old-brokerage	Clear commission overrides
POST	/schemes/update	Update scheme values or launch date
POST	/upload/commission	Upload commission structure workbook
POST	/upload/brokerage-matrix	Upload consolidated brokerage workbook
POST	/upload/launch-dates	Upload launch-date workbook
POST	/old-brokerage/update	Create or update historical data
POST	/old-brokerage/delete	Delete historical data via override/tombstone
POST	/old-brokerage/upload	Upload historical brokerage workbook
GET	/download/demo-commission-file	Download commission template
GET	/old-brokerage/demo-download	Download historical template
POST	/amcs/:amc/documents	Upload up to 20 AMC documents
PUT	/amcs/:amc/documents/:id	Replace an AMC document
DELETE	/amcs/:amc/documents/:id	Delete an AMC document

Admin endpoints

Method	Endpoint	Description
GET	/users	List users
POST	/users	Create a user
PUT	/users/:username/password	Change a user's password
DELETE	/users/:username	Delete a user

All state-changing requests require the CSRF token issued by the application. Uploads are held in memory, limited to 20 MB, and restricted to `.xlsx` / `.xls` ; AMC documents additionally allow `.pdf` .

12. Persistence Layout

Path	Purpose
ter-cache.json	Last successful main scheme snapshot
ter-manual-sync.json	Operator-approved fallback snapshot
commission-override-cache.json	Commission overrides
old-brokerage-override-cache.json	Historical overrides and tombstones
launch-date-override-cache.json	Launch-date overrides
data/app.db	SQLite users database
data/users.json	Legacy user import source
data/amc-documents/<amc-slug>/	AMC document metadata and files

AMC folders contain `documents.json` metadata and a `files/` directory. Scheme and commission data are not stored in SQLite.

13. Security and Operational Notes

- Production startup refuses to run without `SESSION_SECRET` .
- Session cookies are HTTP-only and `SameSite=Lax` .
- Helmet supplies security headers and the application uses a restrictive content security policy.
- Login attempts are limited to 10 per IP per 15 minutes.
- All write requests use double-submit CSRF protection.
- Upload filenames are validated and document files are sanitized before storage.
- The server logs uncaught exceptions and unhandled rejections without silently terminating the process.
- Keep cache, override, SQLite, certificate, and AMC document paths backed up before deployment or bulk uploads.

14. Tests

Run the current Node test suite with:

```
npm test
```

Current test files cover AMC document behavior, cache behavior, user authentication/storage, and year-label handling. Recommended future coverage includes route-level authorization, upload parser fixtures, ratio calculations, refresh fallback behavior, and browser smoke tests for each role.

15. Troubleshooting

Symptom	Checks
Empty or loading matrix	Check server logs, <code>/api/status</code> , AMFI connectivity, and <code>ter-cache.json</code>
Stale data	Check the UI stale marker, AMFI availability, and <code>ter-manual-sync.json</code>
Missing commission	Check workbook headers, scheme-name matching, and commission overrides
Missing historical row	Check ARN, period format, AMC/scheme identity, and tombstones
Upload rejected	Check extension, 20 MB limit, required columns, login, and CSRF token
Login failure	Check <code>data/app.db</code> , credentials, role, and <code>SESSION_SECRET</code>
Document not visible	Check AMC slug, <code>documents.json</code> , and the document <code>files/</code> directory

16. Source Map

Concern	Main files
Bootstrap and middleware	<code>server.js</code> , <code>config/app.js</code>
Pages and views	<code>routes/pages.js</code> , <code>controllers/pageController.js</code> , <code>views/</code>
APIs	<code>routes/api.js</code> , <code>controllers/</code>
TER/NAV ingestion	<code>services/terService.js</code> , <code>services/navService.js</code> , <code>services/terParser.js</code>
Matching	<code>services/matchService.js</code> , <code>helpers/normalizer.js</code>
Refresh and cache	<code>services/dataService.js</code> , <code>services/cacheService.js</code>
Scheduling	<code>services/schedulerService.js</code>
Users/authentication	<code>services/db.js</code> , <code>services/userService.js</code> , <code>middleware/csrf.js</code>
Commission	<code>services/excelUploadService.js</code> , <code>services/commissionOverrideService.js</code>

Concern	Main files
Historical brokerage	<code>services/oldBrokerageService.js</code> , <code>services/Oldbrokerageuploadservice.js</code> , <code>services/oldBrokerageOverrideService.js</code>
AMC documents	<code>services/amcDocumentService.js</code> , <code>controllers/documentController.js</code>
Browser UI	<code>public/js/</code> , <code>public/css/</code> , <code>views/</code>